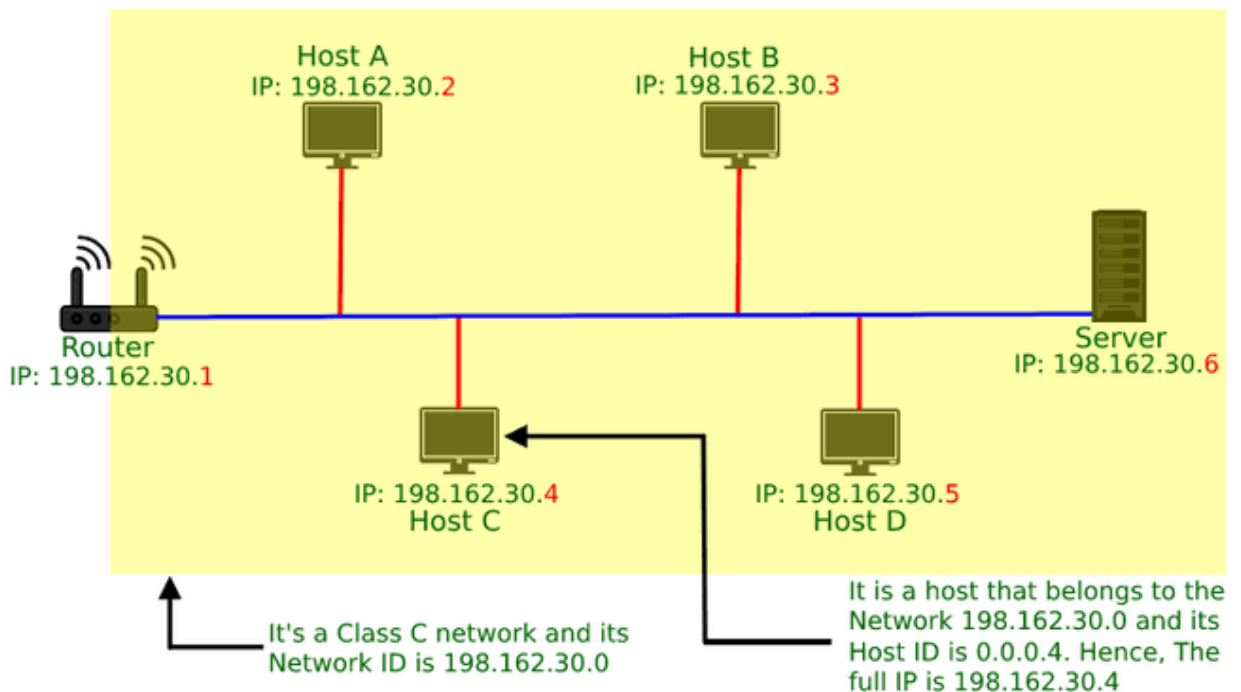


To master advanced networking, you have to look past the decimal numbers you see on your computer screen (192.168.1.1) and look at what the computer actually sees: **a 32-digit string of 1s and 0s.**

An IPv4 address is divided into two distinct parts:

1. **The Network ID:** The "street name" that tells routers which neighborhood your device is in.
2. **The Host ID:** The unique "house number" identifying your exact device on that street
- 3.



1. The Anatomy of an Octet

An IP address is made of four numbers separated by dots. Each of these numbers is called an **octet** because it represents exactly **8 bits** in binary.

Inside an octet, eight binary slots are assigned specific values from right to left based on base-2 exponents: **128, 64, 32, 16, 8, 4, 2, 1.**

If a slot is a **1**, you count its value. If it is a **0**, you ignore it.

- Decimal **192** in binary becomes: **11000000** ($128 + 64 = 192$)
- Decimal **168** in binary becomes: **10101000** ($128 + 32 + 8 = 168$)

2. What is a Subnet Mask?

A **Subnet Mask** is a filter that tells the computer exactly where the Network ID ends and the Host ID begins.

A subnet mask is written just like an IP address, but it uses a solid wall of continuous **1s** followed by continuous **0s**. Wherever the **1s** stop, the network boundary is drawn.

You will often see this written in shorthand as **CIDR notation** (Classless Inter-Domain Routing), using a forward slash like `/24`. The number simply tells you how many **1s** are in the mask.

- **Shorthand:** `/24`
- **Binary Mask:** `11111111.11111111.11111111.00000000`
- **Decimal Mask:** `255.255.255.0`
- *Meaning:* The first 34 bits (the first three numbers) are the unchangeable street name. The last 8 bits belong to the houses.

3. How to Calculate Subnets from Scratch

Imagine you are a network engineer for a company. You are given one standard network block: `192.168.1.0/24`. This gives you 1 neighborhood with 256 total house numbers.

However, you need to isolate **four distinct departments** (HR, Engineering, Sales, and Finance) so their traffic doesn't mix. You must split this single block into 4 smaller subnets.

Here is the exact mathematical procedure to do it:

1. Determine how many bits to borrow: Step 1.

To create more networks, you must steal slots from the Host ID section and give them to the Network ID section.

Use the formula

$$2^N = \text{number of subnets}$$

, where N is the number of bits you need to borrow.

Since you need 4 subnets, and

$$2^2 = 4$$

, you must borrow exactly **2 bits**.

2. Calculate the new Subnet Mask: Step 2.

Your old network mask was a `/24`. You just borrowed 2 bits, so your new mask is a `/26` ($24 + 2$).

Let's look at the last octet in binary now. It used to be all zeros (`00000000`). Now, the first two slots are locked into the network mask (`11000000`).

Add up the decimal values of those two slots: $128 + 64 = 192$.

Your brand-new subnet mask is `255.255.255.192`.

3. Find the Magic Number (Block Size): Step 3.

To find out how large each neighborhood block is, subtract the last non-zero number of your subnet mask from the number 256.

$$\text{Magic Number} = 256 - 192 = 64$$

This tells you that your subnets will step forward by exactly **64 addresses** each time.

4. Map out the Subnet boundaries: Step 4.

Using your magic number of 64, start at 0 and lay out your 4 distinct department networks:

- **Subnet 1 (HR):** Starts at `192.168.1.0`
-
- **Subnet 2 (Engineering):** Starts at `192.168.1.64`
-
- **Subnet 3 (Sales):** Starts at `192.168.1.128`
-
- **Subnet 4 (Finance):** Starts at `192.168.1.192`
-

5. Subtract the un-usable addresses: Step 5.

Every network block loses two addresses to administrative rules:

1. The absolute **first address** is reserved as the Network ID itself.
- 2.
3. The absolute **last address** is reserved as the Broadcast ID (used to yell an announcement to every machine on that specific subnet at once).
- 4.

For Subnet 2 (192.168.1.64):

- **Network Address:** 192.168.1.64 (Unusable for computers)
-
- **First Usable IP:** 192.168.1.65
-
- **Last Usable IP:** 192.168.1.126
-
- **Broadcast Address:** 192.168.1.127 (Unusable for computers)
-

This leaves you with exactly 62 usable host addresses per department network.

Interactive Subnet Experiment

To see how changing the number of borrowed bits dynamically reshapes a network, try moving the slider below. Watch how increasing the CIDR value shrinks the number of usable hosts while multiplying the available network zones.